Juan David Munoz Bolanos

Junior Optical System Engineer

Rechenauerstraße 42, Innsbruck, Austria

+43 676 9115145
jmunozbolanos@gmail.com

Summary

Physicist and Optical Systems Engineer with extensive experience in the end to Polishing complex optomechatronic and medical devices from concept to series. Proven track record in translating physical concepts into robust prototypes using SolidWorks, structured V model systems engineering, and rigorous testing protocols. Adept at collaborating in interdisciplinary teams to deliver high quality technical documentation and system integration for clinical and industrial applications.

Research/Professional Experience

Jan 2022 to Jun 2026 — **Medical University of Innsbruck** — *Innsbruck, Austria*

Optical Systems Engineer | PhD Researcher

End to End System Design: Led the complete hardware to software build of a multiphoton microscope, integrating laser sources, galvo scanners, and adaptive optics. Achieved diffraction limited resolution.

Systems Engineering: Structured instrument development following V model principles, managing requirements tracking from concept to component testing. Utilized FMEA and 8D problem solving to prevent and resolve technical defects.

Quality & Testing: Executed structured Design of Experiments (DoE) and optical simulations in Python. Applied testing and documentation protocols for optomechanical assemblies with awareness of ISO 9001 and ISO 10110 frameworks.

CAD & Prototyping: Designed custom optomechanical mounts and alignment tools using SolidWorks and FreeCAD, ensuring precise mechanical tolerances and seamless integration.

Mar 2020 to Dec 2021 — **Leibniz Institute of Photonic Technology** — *Jena, Germany*

R&D Research Assistant

Spectrometer Prototyping: Designed and built a dispersive 1064 nm fiber probe Raman spectrometer for non invasive human tissue characterization, handling optical layout and detector calibration.

Data Analysis & ML: Developed a custom Python package (SpectraMap) to apply supervised machine learning algorithms to 2D hyperspectral Raman images, achieving high accuracy classification of tissue samples.

Technical Documentation: Authored comprehensive technical reports, system manuals, and peer reviewed publications detailing optical design parameters and calibration procedures.

Technical Competencies

Optics & Hardware: Adaptive optics · Multiphoton/confocal microscopy · Raman spectroscopy · Laser systems (fs, CW, 1064 nm) · Camera integration · 3D CAD (SolidWorks, FreeCAD)

Software & Tools: Python (JAX, scikit-learn, NumPy) · LabVIEW/FPGA · C++ · OpenCV · Git · MATLAB

Soft Skills: Interdisciplinary teamwork · Fast learner on new toolchains

** Dark tags: expertise; light tags: knowledge.*

Education

Jan 2022 to Jun 2026 — **Medical University of Innsbruck** — *Innsbruck, Austria*
Doctor of Philosophy in Optical Metrology

Oct 2019 to Sep 2021 — **Friedrich Schiller Univ. Jena** — *Jena, Germany*
Master of Science in Photonics

Aug 2012 to Aug 2018 — **University of Cauca** — *Popayán, Colombia*
Bachelor of Physics Engineering

Publications

Muñoz Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* (2025).

Muñoz Bolaños, J. D. et al. Dispersive 1064 nm fiber probe Raman. *Applied Sciences* (2024).

Sohmen, M., Muñoz Bolaños, J. D. et al. Optofluidic Adaptive Optics in multiphoton microscopy. *Biomedical Optics Express* (2023).

Soft Skills

Systems & Quality Engineering: V-Model, 8D Problem Solving, Design of Experiments, ISO 9001.

Adaptability & Independent Working: Self-driven approach to bringing concepts to working systems; comfortable in dynamic, fast-paced environments.

Teamwork & Communication: Cross-functional collaboration across optics, hardware, software, and production teams; structured root cause analysis.

Personal Interests

Biking/Hiking | Bachata Dancing | Harmonica | Learning German

Languages

English (Fluent) | German (Intermediate B2) | Spanish (Native)

References

Prof. Dr. Monika Ritsch-Marte
Medical University of Innsbruck
monika.ritsch-marte@i-med.ac.at

PD Dr. Christoph Krafft
Leibniz IPHT
christoph.krafft@leibniz-ipht.de

INTECOL
OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos ($1,500,000) plus three hundred thousand pesos ($300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL
NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO
Administrative and Financial Director

UNIVERSIDAD DEL CAUCA
ON BEHALF OF THE
REPUBLIC OF COLOMBIA
AN BY AUTHORIZATION OF THE MINISTRY OF NATIONAL EDUCATION
CONSIDERING THAT

JUAN DAVID MUÑOZ BOLAÑOS
CC 1.061.772.278 OF POPAYAN

HAS COMPLIED WITH ALL THE STATUTORY AND LEGAL REQUIREMENTS, GRANTS THE TITLE OF

PHYSICAL ENGINEER

WITH ALL THE RIGHTS, PRIVILEGES AND DIGNITIES THAT AUTHORIZE HIM FOR THE PROFESSIONAL EXERCISE

POPAYAN, 16 MARCH 2018

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RESOLUTION No. R-266-07 MINUTE No. 14-07

(SIGNED)

THE PRINCIPAL OF THE UNIVERSITY
THE DEAN OF THE FACULTY
THE SECRETARY GENERAL OF THE UNIVERSITY

Juan Carlos Ojeda - Garces
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RESOLUCIÓN No 1171
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FRIEDRICH-SCHILLER-UNIVERSITÄT JENA
PHYSIKALISCH-ASTRONOMISCHE FAKULTÄT

URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021

Friedrich-Schiller Universität Jena
2

Prof. Dr. Christian Spielmann	Prof. Dr. Silvana Botti
Dekan	*Vorsitzende des Prüfungsausschusses*

FRIEDRICH-SCHILLER-
UNIVERSITÄT
JENA

Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994
in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden

österreich ösd INNSBRUCK schweiz deutschland

Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA

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Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994
in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum
BFI Tirol Bildungs GmbH
in Innsbruck / Österreich

bestanden

Datum: 11.08.2025

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NACHNAMEN Vornamen / SURNAMES Forenames

MUÑOZ BOLAÑOS

Juan David

GESCHLECHT / SEX STAATSANGEHÖRIGKEIT / NATIONALITY GEBURTSDATUM / DATE OF BIRTH

M COL 04 06 1994

KARTE GÜLTIG BIS / CARD EXPIRY

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LA CRUZ

STAATSANGEHÖRIGKEIT / NATIONALITY

KOLUMBIEN

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TITEL VORGESTELLT / TITLE TITEL NACHGESTELLT / POSTNOMINAL TITLE

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OESTERREICH